

IBM Data Analyst Capstone: 2026 Technology Trend and Job Market Insight

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OUTLINE



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EXECUTIVE SUMMARY



Summarize key findings here

- Market Insight: With 5,316 current job postings, Washington, DC is the top IT hub.
- Salary Insight: At \$130,801 annually, Swift is the highest-paying language.
- Database Dominance: The industry favourite, PostgreSQL, is expected to increase significantly in the upcoming year.
- Future Shift: Rust, Go, and TypeScript are becoming more popular among developers.



EXECUTIVE SUMMARY

- **A Mature Workforce:** Since more than 70% of respondents are in the prime professional age range of 25 to 44, the insights are guaranteed to represent the viewpoints of seasoned industry professionals.
- **Highly Educated Talent:** Most employees have a bachelor's or master's degree, demonstrating a high degree of formal technical training.
- **Diverse Tech Roles:** A variety of developers work at Insights, with a primary focus on front-end, back-end, and full-stack development in the financial and technology industries.

INTRODUCTION



- This report using the data analytics to assess the current and projected trends skills needs in relation to programming languages, platforms and databases.
- This report is targeted to mainly IT professionals, Human resources and tech stakeholders seeking data-driven insights into high-demand skills that will remain relevant and future-proof.
- The report answered the questions below:
 - Which programming languages are more popular in today's market?
 - Which IDEs and web frameworks dominate the current market
 - What are the most in-demand database skills required by the industry.



METHODOLOGY



- Data sources, data gathering, and data manipulation procedures
- **Data Gathering**
 - Job Postings: Gathered using a Flask-based API that retrieves technical job data held on IBM Skills Network Cloud Storage (S3) and originally received from Naukri.com.
 - Salary Benchmarks: Taken from a specific HTML dataset, `Programming_Languages.html`, via Web Scraping with BeautifulSoup and Requests.
 - Technology Trends: A thorough subset of the Stack was processed.
 - To find new trends for 2026, use the Overflow Developer Survey.
- Data Organization and Cleaning
 - Standardisation: 18,845 raw records were cleaned using Python (Pandas).
 - Integrity: Strict cleaning was carried out by eliminating duplicate entries and addressing missing values (NaN) in crucial columns (such as Languages, Databases, and Compensation).
 - Profiling: In-depth profiling of developer demographics and key tech stack utilisation was conducted using SQL (SQLite).

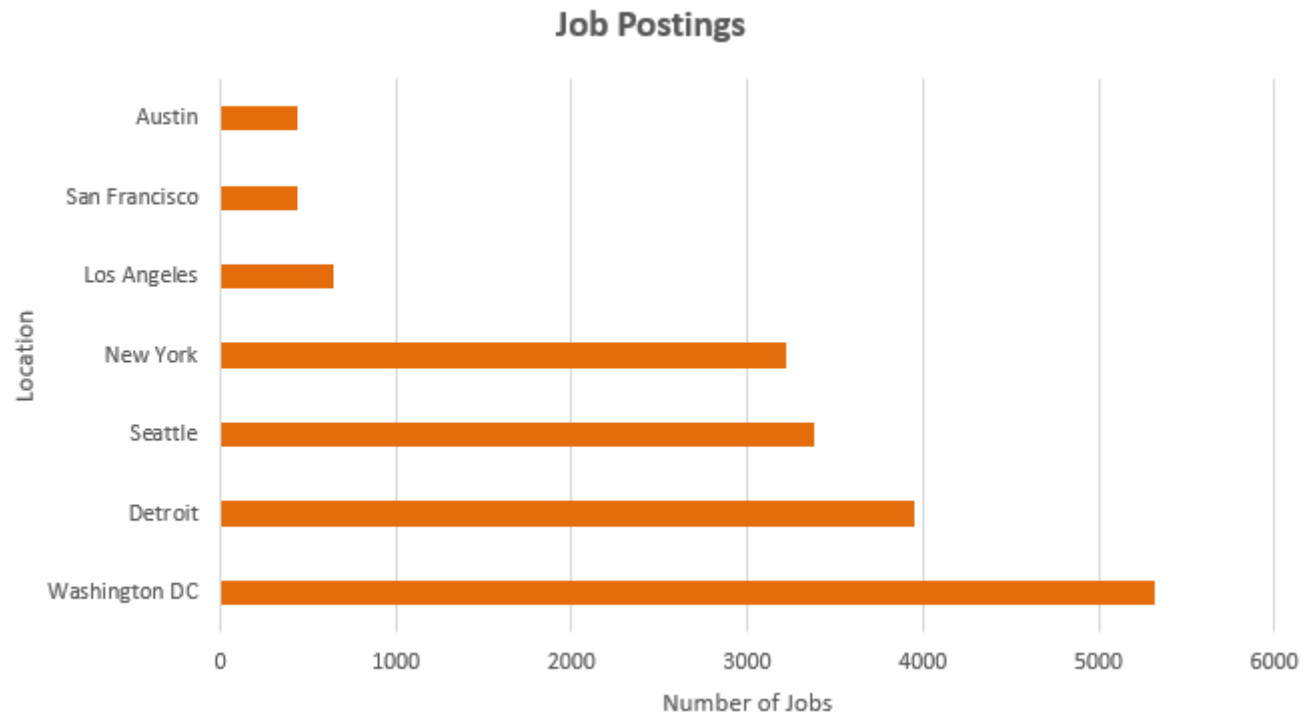


Results



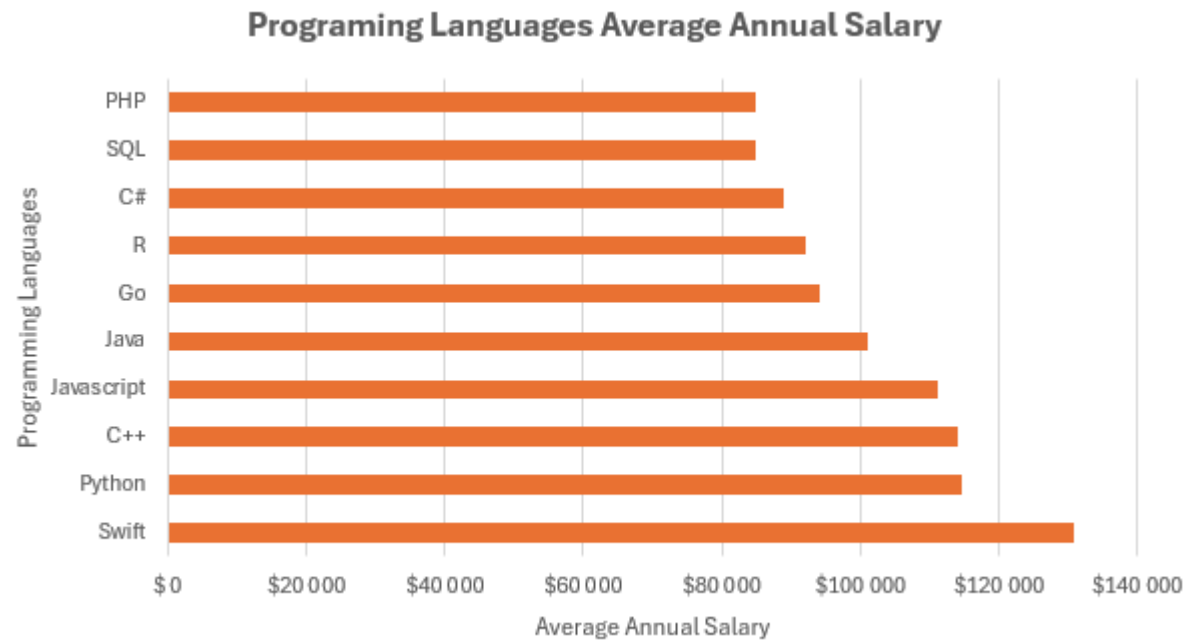
JOB POSTINGS

Washington DC has the highest number of Job Post (5316) while Austin has the least number of Job Post(434)



POPULAR LANGUAGES

Both Swift and Python are the most paying programming languages

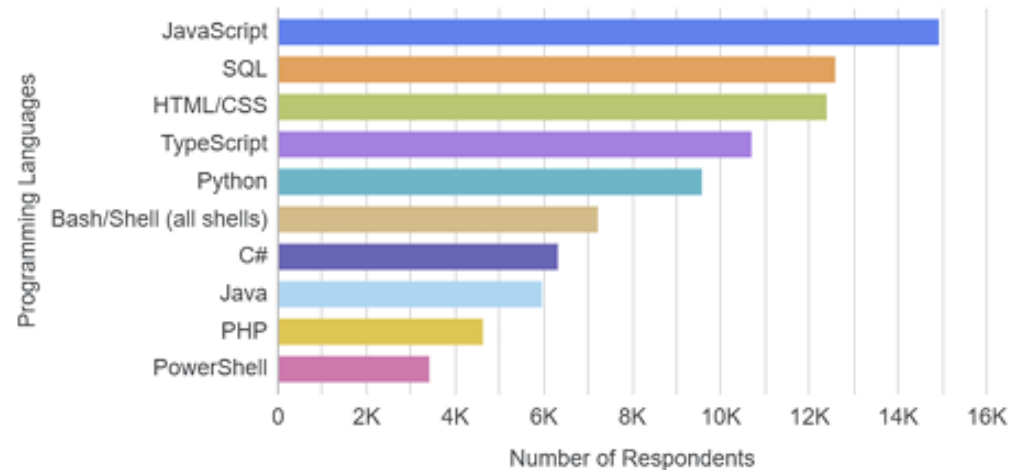


PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

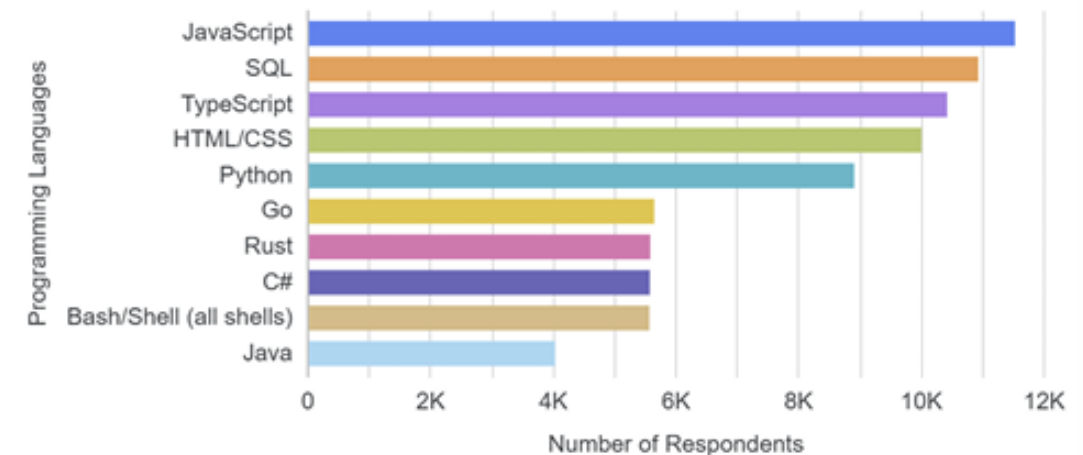
Current Year

Top 10 Programming Languages Used



Next Year

Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript and SQL maintain undisputed dominance in active engineering workflows, anchoring the industry as the baseline core stack.
- Although trailing in immediate raw usage, TypeScript dominates the "Desired Next Year" predictive metrics, signalling an industry-wide migration toward type-safe development architectures.
- Next-generation languages like Go and Rust are capturing substantial market momentum, driven by an escalating architectural demand for high performance and optimal memory efficiency.

Implications

- Software engineers must proactively master TypeScript and Python to maintain market competitiveness, as these dual ecosystems effectively bridge current employment demands with future technological architectures.
- Enterprise organizations must strategically pivot toward next-generation web ecosystems like Next.js and Node.js to align with engineering preferences, optimize product workflows, and secure premium technical talent.
- The surging market momentum behind Go and Rust serves as a clear indicator that companies should evaluate these specific languages for building robust, high-concurrency systems and performance-critical infrastructure.

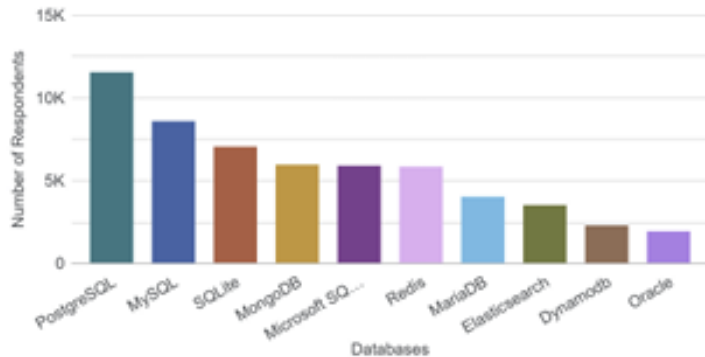


DATABASE TRENDS

Summarize key trends shown in the charts.

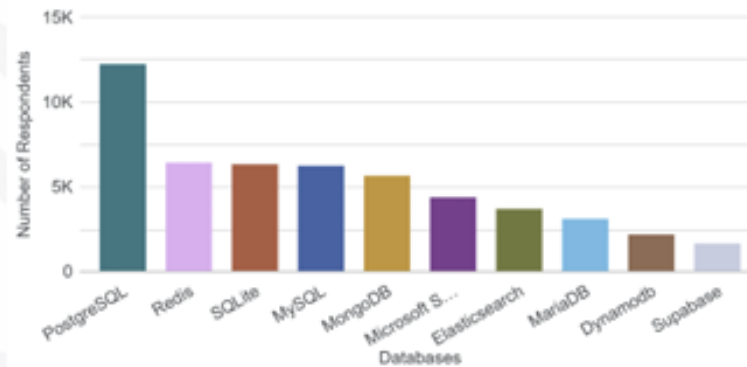
Current Year

Top 10 Databases Used



Next Year

Top 10 Databases Desired Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL Dominance: PostgreSQL is the undisputed leader in the database ecosystem, ranking #1 in both current usage and future developer desire.
- Relational Resilience: Despite the rise of NoSQL, Relational Databases (SQL-based) like PostgreSQL and MySQL continue to be the primary backbone of global data architecture.
- Cloud Alignment: The high usage of AWS, Azure, and Google Cloud directly correlates with the widespread deployment of managed relational database services.

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Implications

- **Architecture Standardization:** PostgreSQL should be the Gold Standard choice for backend architecture due to its stability, feature set, and high community support.
- **Legacy Modernization:** Organizations still relying on older database versions should consider migration to PostgreSQL to stay aligned with the modern talent pool's expertise.
- **Core Competency:** SQL proficiency remains a non-negotiable skill for 2026, regardless of the emerging technologies in other layers of the stack.

DASHBOARD



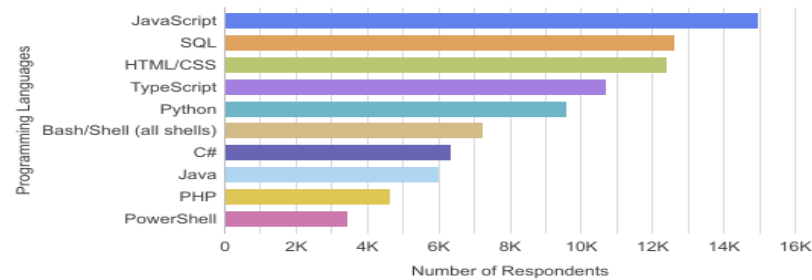
<Please present your dashboard in the following slides.>



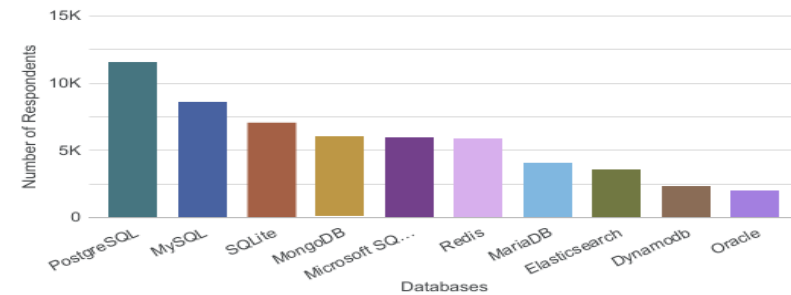
DASHBOARD TAB 1: Current Technology Usage

Current Technology Usage

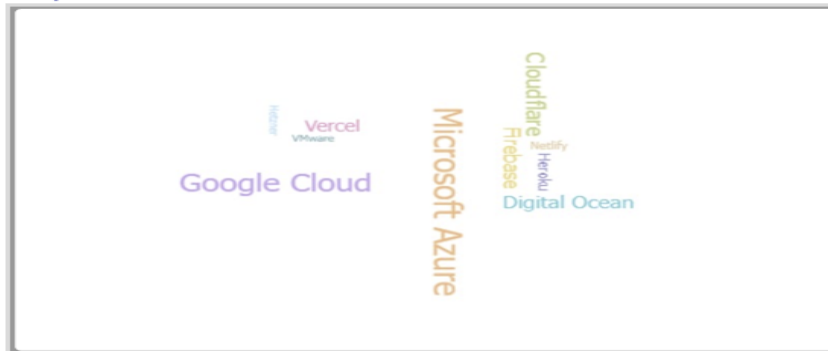
Top 10 Programming Languages Used



Top 10 Databases Used



Top 10 Platforms Used



Top 10 Web Frameworks



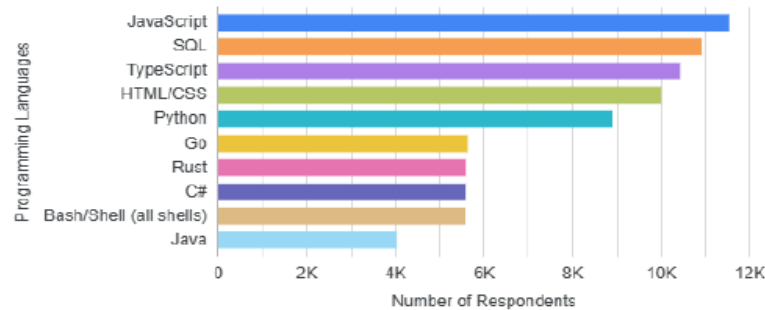
DASHBOARD TAB 1: Current Technology Usage

- **Foundational Stack:** As the foundation of both online and data infrastructures, JavaScript and SQL continue to be the unquestionable industry pillars.
- **Database Standard:** With their high dependability for enterprise systems, PostgreSQL and MySQL rule the relational database market.
- **Cloud Dominance:** The Big Three providers—AWS, Azure, and Google Cloud—dominate the current market, demonstrating a strong reliance on significant cloud infrastructures.
- **Web Framework Stability:** The most popular frameworks are Node.js and React, while jQuery is still widely used, suggesting a combination of contemporary and traditional web technologies.
- **Understanding:** In order to preserve the current global digital underpinnings, the market nevertheless requires proficiency with conventional, reliable technology.

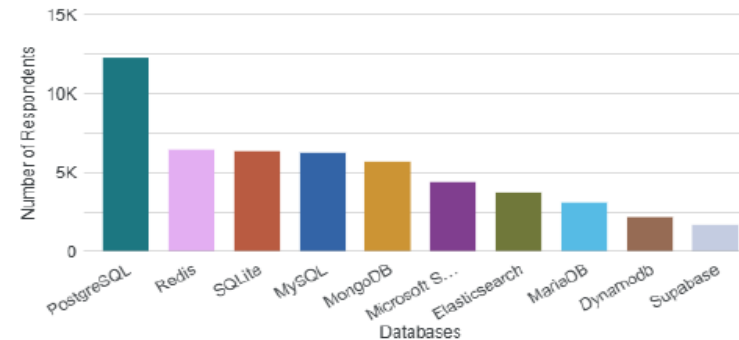
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trends

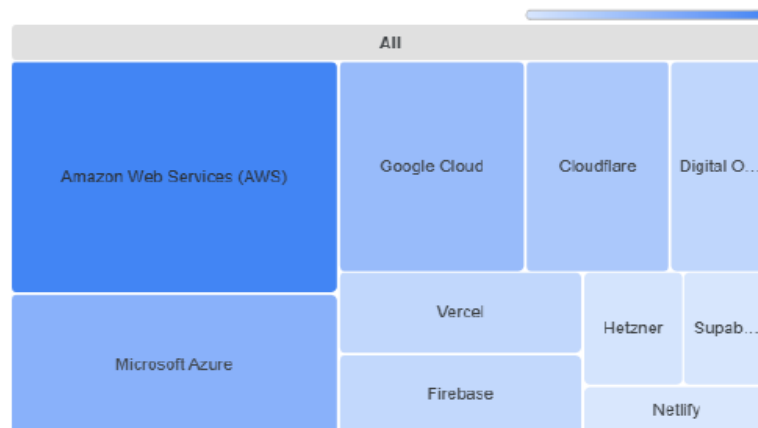
Top 10 Languages Desired Next Year



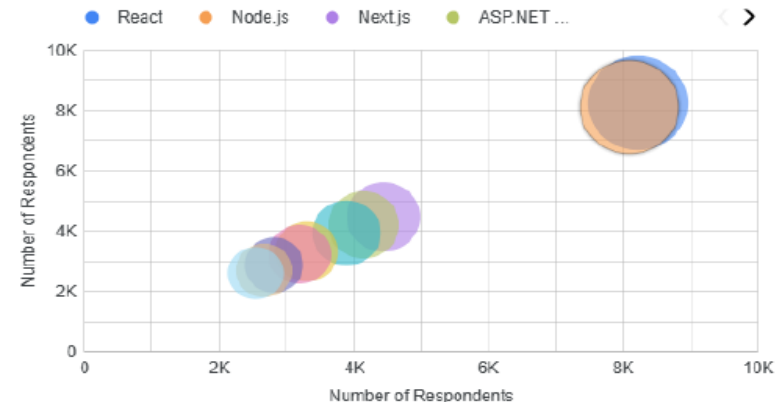
Top 10 Databases Desired Next Year



Top 10 Desired Platforms



Top 10 Desired Web Frameworks



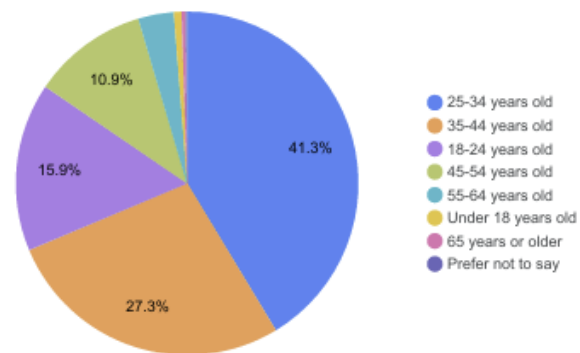
DASHBOARD TAB 2: Future Technology Trends

- **The Shift:** The necessity for type-safety and high-performance execution is driving a large developer migration toward TypeScript, Go, and Rust.
- **Platform Consistency:** In order to maintain their market hegemony for the next year, developers plan to stick with the major providers (AWS, Azure, and Google Cloud).
- **Framework Evolution:** Next.js's rise to the top ranks indicates a distinct trend toward server-side rendering and contemporary React ecosystems, even tho React and Node.js continue to be top priorities.
- **Insight:** To draw in top people and keep a competitive edge, organisations need to get ready for a "Modern Stack" transition.

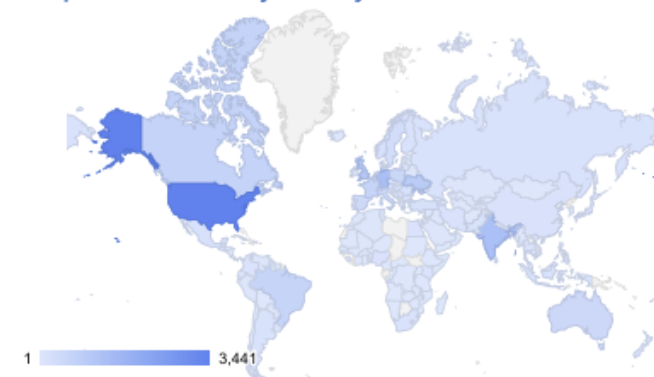
DASHBOARD TAB 3: Demographics

Demographics

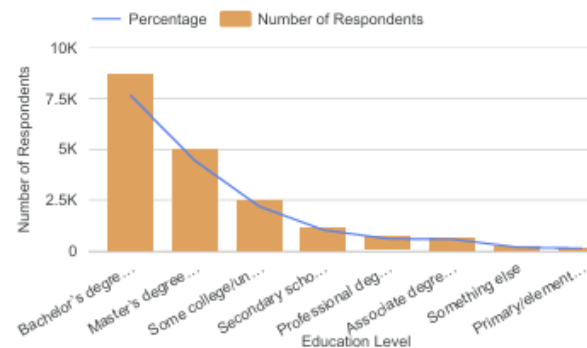
Respondents by Age



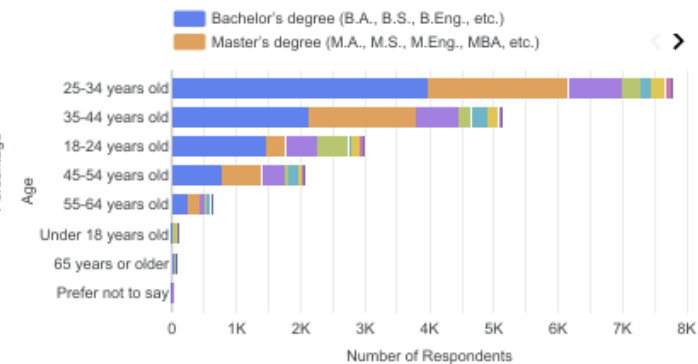
Respondents Count by Country



Respondent Distribution by Education Level



Respondent Count by Age, Classified by Education Level



DASHBOARD TAB 3: Demographics

- Experience Pool: Mature professionals between the ages of 25 and 44 make up the majority of the workforce, which is a highly productive and experienced talent pool.
- Education: A solid academic background—mostly a bachelor's or master's degree—remains a crucial prerequisite for employment in the tech sector.
- Geographic Hubs: The USA, Germany, and India, which comprise the largest talent pools in this analysis, are the three main tech countries that drive the survey data.

DISCUSSION



OVERALL FINDINGS & IMPLICATIONS

Findings

- **Market Dominance:** Swift is the highest-paying expertise (\$130,801), and Washington, DC, is the leading job hub (5,316 listings). **Cloud Monopoly:** For both present and future platforms, AWS, Azure, and Google Cloud continue to be the clear leaders.
- **Web Tech Shift:** There is a strong future migration tendency toward Next.js, yet React and Node.js are highly stable.
- **Global Talent:** The USA, Germany, and India have the biggest concentrations of developers.

OVERALL FINDINGS & IMPLICATIONS

Implications

- Strategic Sourcing: Concentrate on hiring and professional development in Washington, DC, and top talent countries (USA, Germany, India).
- Essential Skills: By 2026, mastery of the Big Three cloud platforms will be required.
- Stack Modernisation: In order to draw and keep top talent, organisations need to switch from old tools to Next.js.
- High-ROI Specialisation: To optimise earning potential, developers should give priority to Python or Swift.

CONCLUSION



An overview of the technology landscape in 2026

- The Cloud Monopoly: 2026 is referred to as the "Cloud-Native Era." For both present operations and upcoming migrations, AWS, Azure, and Google Cloud continue to be the unchallenged infrastructure standards.
- The Relational Backbone: PostgreSQL is the most widely used database standard, both now and in the future, according to developers.
- Web Framework Evolution: Modern, high-performance ecosystems like React, Node.js, and particularly Next.js are replacing outdated tools like jQuery.
- High-Value Talent: Python continues to offer the finest combination of high compensation and widespread demand, while specialised languages like Swift continue to offer the most profitable chances.

CONCLUSION

Strategic Suggestions

1. For Upskilling Developers:

- Put the Modern Stack First: Learn Next.js and TypeScript to keep up with the industry's move away from outdated web frameworks.
- Cloud Proficiency: Since AWS and Azure dominate worldwide infrastructure, obtain certifications or in-depth knowledge of at least one major provider.

2. Regarding Organisations (Modernisation):

- Standardise on PostgreSQL: Match backend architecture to the community's most popular and reliable database.
- Optimise for Performance: To satisfy the increasing need for server-side rendering and an improved developer experience, update web apps with Next.js.

3. Regarding Hiring (Strategic Sourcing):

- Since the USA, Germany, and India will have the biggest and busiest developer populations in 2026, concentrate talent acquisition efforts there.



APPENDIX



- Primary Sources of Information
 - Data from the Stack Overflow Developer Survey Subset (License: ODbL).
 - IBM Cloud Object Storage (S3) is the job API source; the dataset is based on listings on Naukri.com.
- Source for Web Scraping: https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DA0321EN-SkillsNetwork/labs/datasets/Programming_Languages.html
- Stack of Technology
 - Python and SQL are two programming languages.
 - Python Libraries:
 - Data Organization: NumPy and Pandas
 - Web scraping: Requests, BeautifulSoup
 - Visualisation: Plotly, Seaborn, and Matplotlib
- SQLite (Relational EDA) is the database engine.
- Google Looker Studio (Interactive Dashboards), a BI tool

